

1. Sum of digit ||

PROBLEMS

SUBMIT CODE

MY SUBMISSIONS

STATUS

STANDINGS

CUSTOM INVOCATION

E. Sum of Digits - II

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a very large non-negative integer N that may not fit in 64-bit integer types.

Your task is to compute the sum of its digits.

Input

The only line contains the integer N as a string.

It is guaranteed that $1 \leq |N| \leq 10^5$ and N consists only of digits 0–9.

Output

Print one integer — the sum of the digits of N .

Example

input	Copy
1000000000000000000000000	
output	Copy
4	

100xSchool I

Prog

33

Solved 46

Atter

Recommend

Problem

[Hello World](#)
[Print First 5 Alphabets](#)
[Triangle](#)
[Print Z](#)
[Print Table of 5](#)
[Rectangle](#)

If digit is more than int size and long long size

```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 // int -> 10^9
5
6 int main()
7 {
8     int n;
9     cin >> n;
10
11     int sum = 0;
12     while(n != 0)
13     {
14         sum += n % 10;
15         n /= 10;
16     }
17
18     cout << sum << endl;
19 }
```

test 0 edit
time: 146ms
1000000000
1
accept

test 1 edit run
time: 6ms
7253
17
accept
next test

dhruv


```

3
4 // int -> 10^9
5 // long long -> 10^18
6
7 int main()
8 {
9     string s;
10    cin >> s;
11
12    int n = s.size(), sum = 0;
13
14    // s = "7253"
15
16    // s[0] -> '7'
17    // s[1] -> '2'
18    // s[2] -> '5'
19    // s[3] -> '3'
20
21    for(int i = 0; i < n; i++)
22    {
23        sum += s[i];
24    }
25

```

test 0 edit run time: 533ms

7253

209

accept

test 1 edit run time: 9s

1

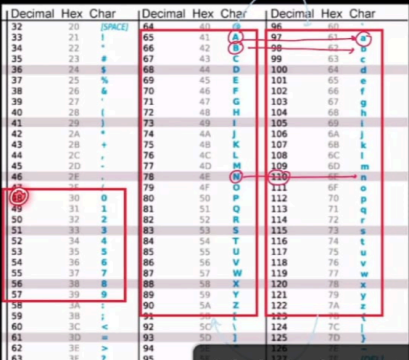
49

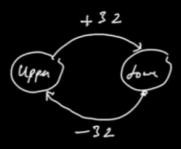
accept

next test



dhruv





```

char ch;
cin >> ch;

if (ch >= 'a' and ch <= 'z')
{
    cout << "lower"
}
else if (ch >= 'A' and ch <= 'Z')
{
    cout << "upper"
}
else if (ch >= '0' and ch <= '9')
{
    cout << "digit"
}

```



dhruv

vishal Biradar to Everyone
 49+7-1
CHARACTER TO ASCII

Very big no's (long long Cant store)

$N \sim 10^9$

Cont << sum << endl;

```
s = "7253"
0123

int n = s.size(), sum = 0;

for (i = 0; i < n; i++)
{
    sum += s[i];
}

cout << sum;
```

48 → 0
49 → 1
50 → 2
51 → 3
52 → 4
53 → 5
54 → 6
55 → 7
56 → 8
57 → 9

ashutosh sao to Everyone
ok thanks

dhruv

```
6 // long long Cant store
7 int main()
8 {
9     string s;
10    cin >> s;
11
12    int n = s.size(), sum = 0;
13
14    // s = "7253"
15
16    // s[0] → '7'
17    // s[1] → '2'
18    // s[2] → '5'
19    // s[3] → '3'
20
21    for (int i = 0; i < n; i++)
22    {
23        sum += s[i] - '0';
24    }
25
26    cout << sum;
27 }
```

test 0 edit run time: 146ms
7253
17
accept

test 1 edit run time: 6ms
1
1
accept

next test

dhruv

2. String Compression

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F. String Compression

time limit per test: 1 second
memory limit per test: 256 megabytes

You have been given a program to perform basic string compression.

For any character that is consecutively repeated more than once, replace the consecutive occurrences with the character followed by the number of repetitions.

For example:

- aaaaaa becomes a6

A character is compressed only when its consecutive count is greater than 1.

Input

The first line contains an integer T ($1 \leq T \leq 5$) — the number of test cases.

Each of the next T lines contains a string S ($1 \leq |S| \leq 5000$).

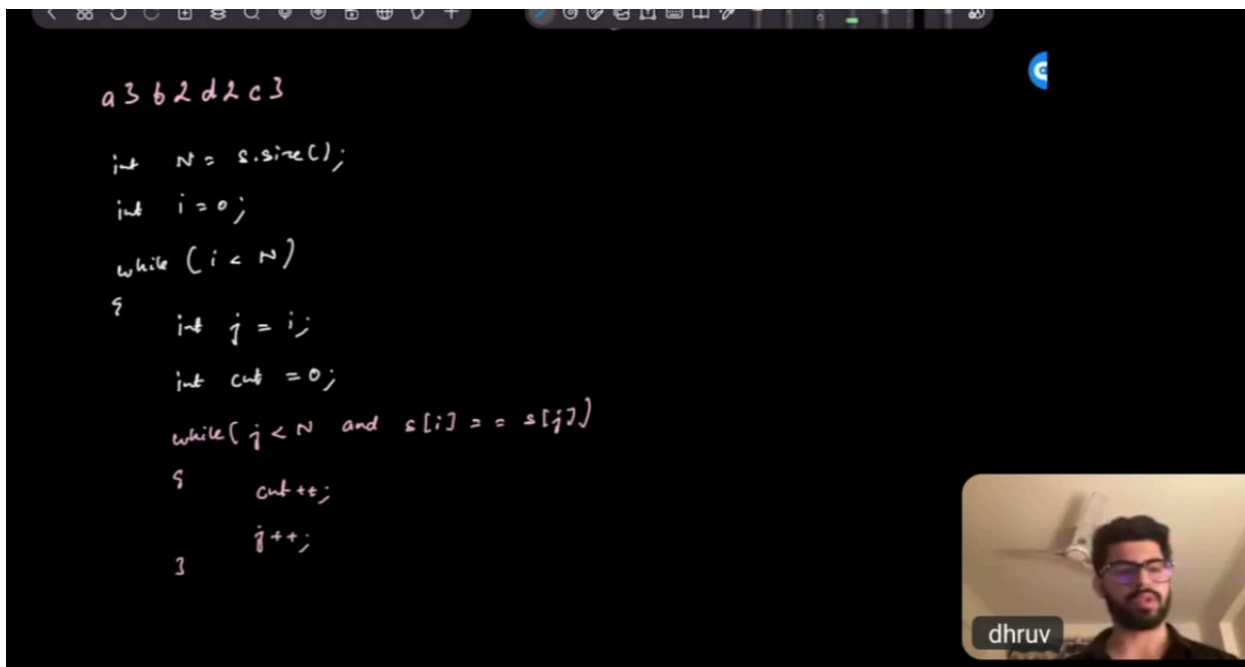
It is guaranteed that the consecutive count of any character in the string does not exceed 9.

Output

For each test case, print the compressed string.

Example

input	Copy
2 aaabddcccc ggttttffffreee	
output	Copy
a3b2d2c3 g2t4f4r2e3	



3. Largest word

Please subscribe to the official Codeforces channel in Telegram via the link https://t.me/codeforces_of

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G. Longest Word

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a string S consisting of English letters and spaces.

A word is a maximal contiguous sequence of letters. It is guaranteed that words are separated by exactly one space, and the first character of S is a letter.

Your task is to find the length of the longest word in S .

Input

The only line contains the string S .

It is guaranteed that $1 \leq |S| \leq 10^5$, and S consists only of English letters and spaces.

Output

Print one integer — the length of the longest word in S .

Example

input	Copy
hello world this is CP	
output	Copy
5	

The image shows a digital whiteboard interface with a dark background. At the top, there's a title "LONGEST WORD" underlined. Below it, there are two examples of input strings and their corresponding output lengths. The first example shows the input "I am learning DSA" and the output "8". The second example shows the input "Improving Myself Every day" and the output "9". In the bottom right corner, there is a small video feed of a person with a beard and glasses, identified by the name "dhruv" below it.

LONGEST WORD

inp: " I am learning DSA "

o/p: 8

inp: " Improving Myself Every day "

o/p: 9

dhruv

9:41 AM Tue 9 Jan

Untitled Notebook 71

STRINGS (1).pdf

```
int N = s.size();
int i = 0;
while (i < N)
{
    if (s[i] == ' ')
    {
        i++;
        continue;
    }

    int j = i;
    int cnt = 0;
    while (j < N and s[j] != ' ')
    {
        cnt++;
        j++;
    }

    if (cnt > ans)
    {
        ans = cnt;
    }
}
```

solve.cpp

```
14 int ans = 0;
15
16 int i = 0;
17 while(i < n)
18 {
19     if(s[i] == ' ')
20     {
21         i++;
22         continue;
23     }
24
25     int j = i;
26     int cnt = 0;
27
28     while(j < n and s[j] != ' ')
29     {
30         cnt++;
31         j++;
32     }
33
34     if(cnt > ans)
35     {
36         ans = cnt;
37     }
38 }
```

solve.cpp -run

test 0 edit run time: 531ms

I am learning c++

a3b2d2c3

accept

test 1 edit run time: 7ms

abbcccd

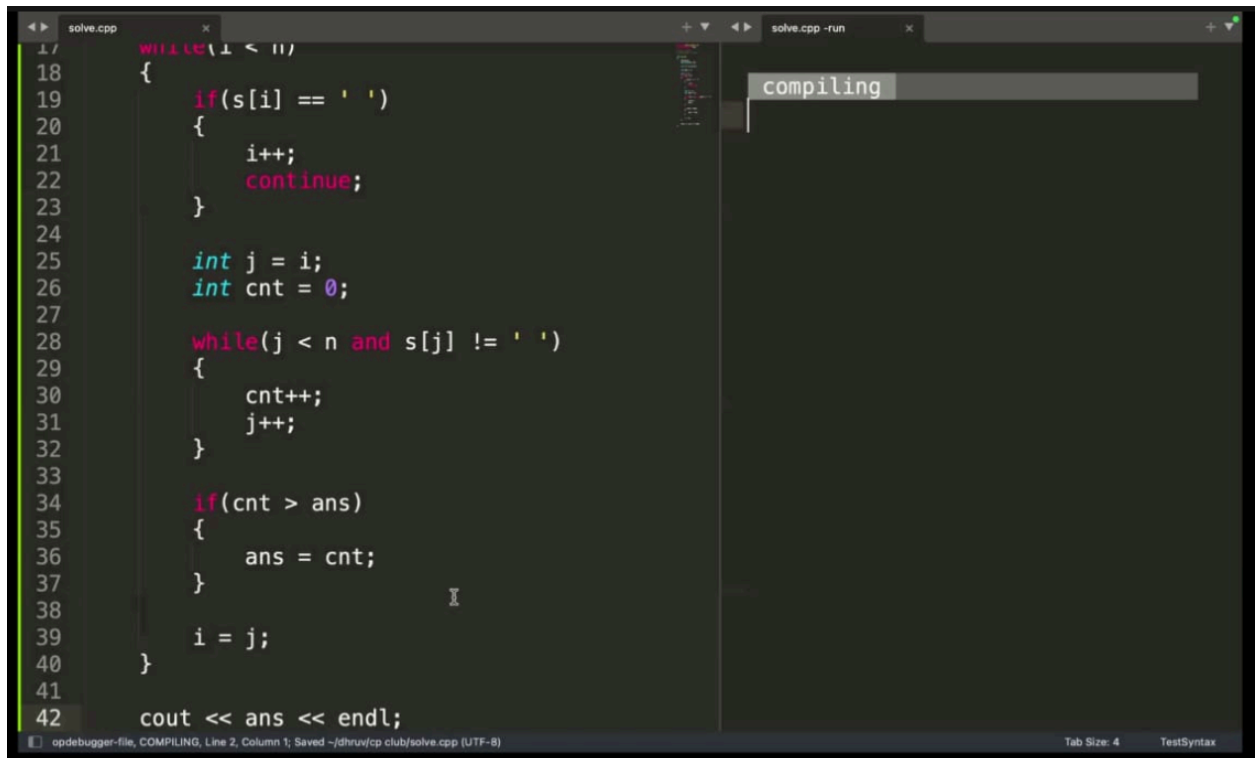
ab2c3d6

accept

next test

Line 34, Column 21

Tab Size: 4 C++



```
17 while(i < n)
18 {
19     if(s[i] == ' ')
20     {
21         i++;
22         continue;
23     }
24
25     int j = i;
26     int cnt = 0;
27
28     while(j < n and s[j] != ' ')
29     {
30         cnt++;
31         j++;
32     }
33
34     if(cnt > ans)
35     {
36         ans = cnt;
37     }
38
39     i = j;
40 }
41
42 cout << ans << endl;
```

compiling

opdebbuger-file, COMPILING, Line 2, Column 1; Saved ~/dhruv/cp club/solve.cpp (UTF-8) Tab Size: 4 TestSyntax

4. Diagonal Sum

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

H. Diagonal Sum

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a square integer matrix of size $n \times n$.

Your task is to compute the sum of the main diagonal (top-left to bottom-right).

Input

The first line contains a single integer n ($1 \leq n \leq 1000$) — the size of the matrix.

The next n lines each contain n integers $a_{i,j}$ ($0 \leq a_{i,j} \leq 10^5$), representing the matrix.

Output

Print a single integer, the sum of the main diagonal elements.

Example

input	Copy
3 1 2 3 4 5 6 7 8 9	
output	Copy
15	

Diagonal Sum

Given a $N \times N$ 2D Array, Find the Sum of Diagonal Elements.

i/p:

5	6	8
2	5	1
9	8	6

o/p: $5 + 5 + 6 = 16$

i/p:

1	2	6	8
3	6	2	5
9	7	8	5
6	6	2	9

o/p: $1 + 6 + 8 + 9 = 24$

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Untitled Notebook 71

STRINGS (1).pdf

Find the sum of diagonal elements.

i/p:

0	1	2
5	6	8
1	2	5
2	9	8

o/p: $5 + 5 + 6 = 16$

(0,0) (1,1) (2,2)

i/p:

0	1	2	3
1	3	6	2
2	9	7	8
3	6	6	2

o/p: $1 + 6 + 8 + 9 = 24$

(0,0) (1,1) (2,2) (3,3)

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STRINGS (1).pdf

Find the sum of diagonal elements.

i/p:

0	1	2
5	6	8
1	2	5
2	9	8

o/p: $5 + 5 + 6 = 16$

(0,0) (1,1) (2,2)

i/p:

0	1	2	3
1	3	6	2
2	9	7	8
3	6	6	2

o/p: $1 + 6 + 8 + 9 = 24$

(0,0) (1,1) (2,2) (3,3)

Row No = Col No

```

int i = 0;
int j = 0;

int sum = 0;
while (i < N and j < N)
{
    sum += A[i][j];
    i++;
    j++;
}
  
```

0 1 2 3

5. Diagonal Sum (Opposite direction)

Untitled Notebook 71

```

while (i < N and j < N)
{
    sum += A[i][j];
    i++;
    j++;
}

```

⇒

```

int sum = 0;
while (i < N)
{
    sum += A[i][i];
    i++;
}

```

$N \times N$

$i = 0;$
 $j = N - 1;$

$i++$

i/p:

0	1	2	3
1	3	6	5
2	9	7	8
3	6	2	9

Akash Jayan to Everyone
i++ j--

Untitled Notebook 71

```

sum += A[i][j];
i++;
j++;
}

```

while (i < N)

```

{
    sum += A[i][i];
    i++;
}

```

$N \times N$

$i = 0;$
 $j = N - 1;$

$i++$
 $j--$

i/p:

0	1	2	3
1	3	6	5
2	9	7	8
3	6	2	9

6. Rotate Image

I. Rotate Image

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a square matrix of size $n \times n$ filled with integers. Your task is to rotate the matrix by 90° in the **clockwise** direction.

Input

The first line contains a single integer n ($1 \leq n \leq 1000$), the size of the matrix.

Each of the next n lines contains n integers $a_{i,j}$ ($1 \leq a_{i,j} \leq 10^9$), the elements of the matrix.

Output

Print the rotated matrix.

Examples

input	Copy
3 1 2 3 4 5 6 7 8 9	
output	Copy
7 4 1 8 5 2 9 6 3	

input	Copy
5 12 47 3 88 65 29 54 76 11 92 7 81 34 56 23 90 14 67 45 38 19 72 8 99 50	
output	Copy
19 90 7 29 12 72 14 81 54 47 8 67 34 76 3 99 45 56 11 88 50 38 23 92 65	

Given a $N \times N$ 2D Array, Rotate it 90° Clock-wise.

i/p:

1	2	3
4	5	6
7	8	9

o/p:

7	4	1
8	5	2
9	6	3

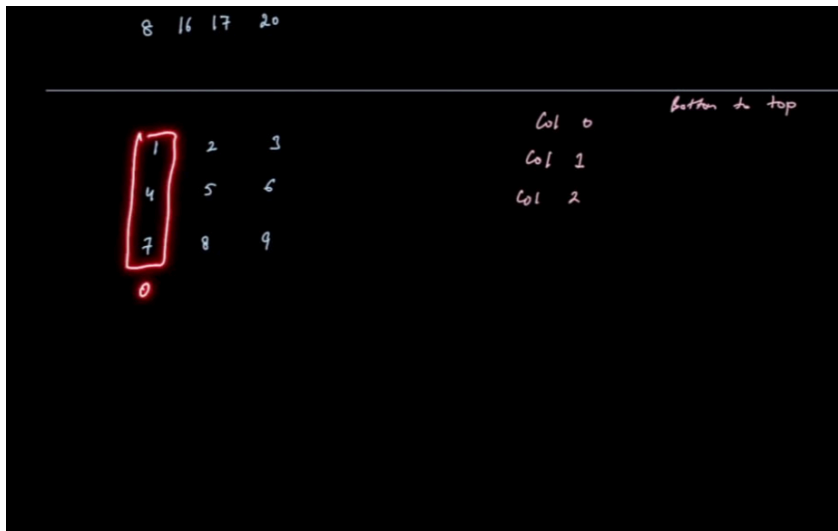
i/p:

1	2	3	4
5	6	7	8
10	5	6	9
8	16	17	20

o/p:

8	10	5	1
16	5	6	2
17	6	7	3
9	8	4	

Sayoun Parui to Everyone
yes



```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 // int -> 10^9
5 // long long -> 10^18
6
7 int main()
8 {
9     int n;
10    cin >> n;
11
12    int arr[n][n];
13    for(int i = 0; i < n; i++)
14    {
15        for(int j = 0; j < n; j++)
16        {
17            cin >> arr[i][j];
18        }
19    }
20
21    for(int j = 0; j < n; j++)
22    {
23        for(int i = n - 1; i >= 0; i--)
24        {
25            cout << arr[i][j] << " ";
26        }
27        cout << endl;
28    }
29 }
```

test 0 edit run time: 579ms

```
3
1 2 3
4 5 6
7 8 9
```

7 4 1
8 5 2
9 6 3

accept

next test

Note: If 90 degree rotation we know then 180 degree 360 degree also similar

7. Spiral Matrix

J. Spiral Matrix

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given an integer matrix with m rows and n columns.

Print all elements of the matrix in **clockwise spiral order** starting from the top-left cell: go right, then down, then left, then up, and repeat while shrinking the remaining rectangle.

Input

The first line contains two integers m and n ($1 \leq m, n \leq 10$).

Each of the next m lines contains n integers, the matrix elements ($-100 \leq a_{i,j} \leq 100$).

Output

Print $m \cdot n$ integers in one line: the elements in spiral order, separated by single spaces.

Examples

input	Copy
3 3 1 2 3 4 5 6 7 8 9	
output	Copy
1 2 3 6 9 8 7 4 5	
input	Copy
3 4 1 2 3 4 5 6 7 8 9 10 11 12	
output	Copy
1 2 3 4 8 12 11 10 9 5 6 7	

100xSchool Bo

Progr

33%

Solved 46 Attempt

Recommended

Problem

[Hello World](#)

[Print First 5 Alphabets](#)

[Triangle](#)

[Print Z](#)

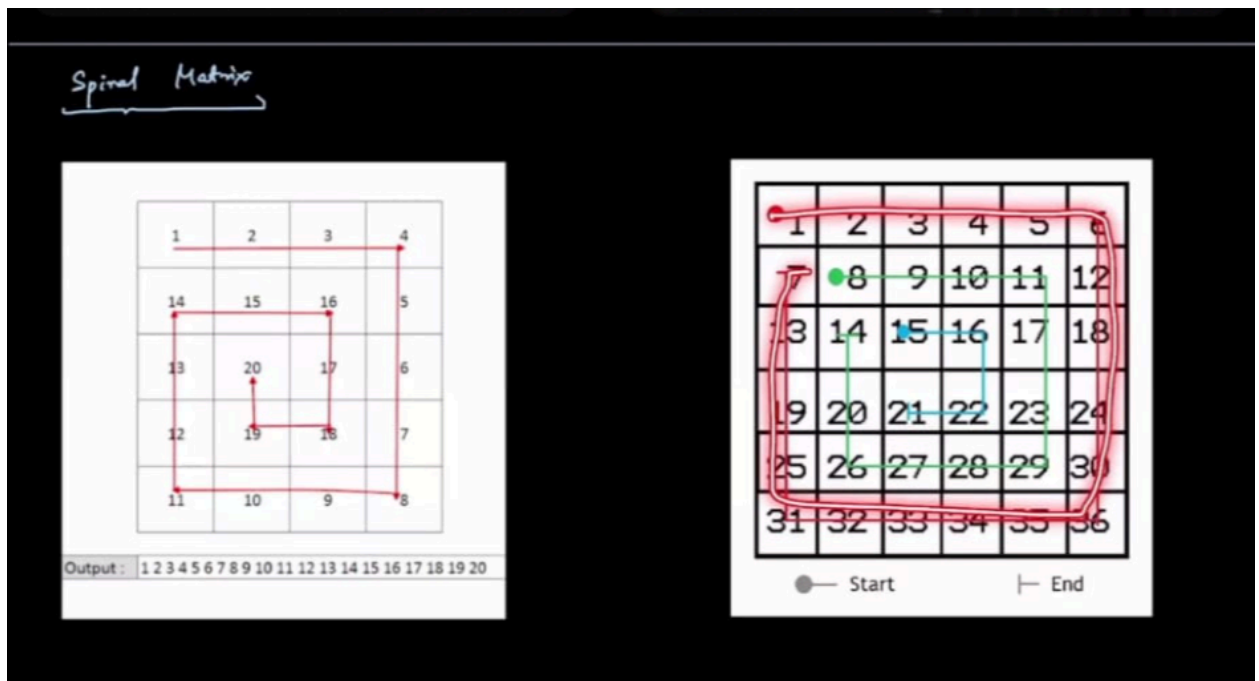
[Print Table of 5](#)

[Rectangle](#)

[Print Table of N](#)

[About Group](#)

Prerequisite : You have to Boundary travel matrix



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STRINGS (1).pdf

Boundary

0	1	2	3	4	5
6	7	8	9	10	
11	12	13	14	15	
16	17	18	19	20	

```
// Start Row ( i = 0 , j → 0 ... M-1 )
for ( j = 0 ; j < M ; j++ )
{
    cout << arr[0][j];
}
```

98% 17/19

```
// End Col ( j = M-1 , i → 1 ... N-1 )
for ( i = 1 ; i < N ; i++ )
{
    cout << arr[i][M-1];
}
```

```
// End Row ( i = N-1 , j → M-2 ... 0 )
for ( j = M-2 ; j >= 0 ; j-- )
{
    cout << arr[N-1][j];
}
```

```
// Start Col ( j = 0 , i → N-2 ... 1 )
for ( i = N-2 ; i >= 1 ; i-- )
{
    cout << arr[i][0];
}
```

ashutosh sao to Everyone
yes

Spiral Matrix

Output: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36

Start End

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STRINGS (1).pdf

100%

StartRow = 0

endRow = N-1

startCol = 0

endCol = M-1

```

// End Row (i = endRow, j → M-2 ... 0)
for(j = M-2; j >= 0; j--)
{
    count << arr[endRow][j];
}
// Start Col (j = 1, i → N-2 ... 1)
for(i = N-2; i >= 1; i--)
{
    count << arr[i][startCol];
}

startRow++;
endRow--;
startCol++;
endCol--;
}

```

Move into the next layer

```

1 #include<bits/stdc++.h>
2 using namespace std;
3
4 // int -> 10^9
5 // long long -> 10^18
6
7 int main()
8 {
9     int n, m;
10    cin >> n >> m;
11
12    int arr[n][m];
13    for(int i = 0; i < n; i++)
14    {
15        for(int j = 0; j < m; j++)
16        {
17            cin >> arr[i][j];
18        }
19    }
20
21    int startRow = 0, endRow = n - 1;
22    int startCol = 0, endCol = m - 1;
23
24    while(startRow <= endRow and startCol <= endCol)
25    {
26        // boundary traversal for this layer
27
28    }
29
30 }

```

test 0 edit run time: 579ms

```

3
1 2 3
4 5 6
7 8 9
7 4 1
8 5 2
9 6 3
accept

```

test 1 edit run time: 15s

```

4
1 2 3 4
5 6 7 8
10 5 6 9
8 16 17 20
8 10 5 1
16 5 6 2
17 6 7 3
20 9 8 4
accept
next test

```

Line 28, Column 9 Tab Size: 4 C++

```
41 solve.cpp
22 int sr = 0, er = n - 1;
23 int sc = 0, ec = m - 1;
24 while(sr <= er and sc <= ec)
25 {
26     // boundary traversal for this layer
27
28     // sr (sc -> ec)
29     for(int j = sc; j <= ec; j++)
30     {
31         cout << arr[sr][j] << " ";
32     }
33
34     // ec (sr + 1 -> er)
35     for(int i = sr + 1; i <= er; i++)
36     {
37         cout << arr[i][ec] << " ";
38     }
39
40     // er (ec - 1 -> sc)
41     for(int j = ec - 1; j >= sc; j--)
42     {
43         cout << arr[er][j] << " ";
44     }
45
46     // sc (er - 1 -> sr + 1)
47     for(int i = er - 1; i >= sr + 1; i--)
48     {
49         cout << arr[i][sc] << " ";
50     }
51
52     sr++;
53     er--;
54
55     sc++;
56     ec--;
57 }
58 }
```

test 0 edit run time: 534ms

```
3 3
1 2 3
4 5 6
7 8 9
```

1 2 3 6 9 8 7 4 5
accept

test 1 edit run time: 1278ms

```
3 4
1 2 3 4
5 6 7 8
9 10 11 12
```

1 2 3 4 8 12 11 10 9 5 6 7 6
accept

test 2 edit run time: 37s

```
6 6
1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20 21 22 23 24
25 26 27 28 29 30
31 32 33 34 35 36
```

1 2 3 4 5 6 12 18 24 30 36 35 34 33 32 31 25 19 13 7 8
9 10 11 17 23 29 28 27 26 20 14 15 16 22 21
accept

test 3 stop

```
1 5
1 2 3 4 5
```

opendevolver-file, RUNNING, 9 characters selected

Tab Size: 4 TestSyntax

```
32 // boundary traversal for this layer
33
34 // sr (sc -> ec)
35 for(int j = sc; j <= ec; j++)
36 {
37     cout << arr[sr][j] << " ";
38 }
39
40 // ec (sr + 1 -> er)
41 for(int i = sr + 1; i <= er; i++)
42 {
43     cout << arr[i][ec] << " ";
44 }
45
46 if(sr != er)
47 {
48     // er (ec - 1 -> sc)
49     for(int j = ec - 1; j >= sc; j--)
50     {
51         cout << arr[er][j] << " ";
52     }
53 }
54
55 if(sc != ec)
56 {
57     // sc (er - 1 -> sr + 1)
58     for(int i = er - 1; i >= sr + 1; i--)
59     {
60         cout << arr[i][sc] << " ";
61     }
62 }
63
64 sr++;
65 er--;
66
67 sc++;
68 ec--;
```

9 10 11 12

1 2 3 4 8 12 11 10 9 5 6 7 6
accept

test 2 edit run time: 37s

```
6 6
1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20 21 22 23 24
25 26 27 28 29 30
31 32 33 34 35 36
```

1 2 3 4 5 6 12 18 24 30 36 35 34 33 32 31 25 19 13 7 8
9 10 11 17 23 29 28 27 26 20 14 15 16 22 21
accept

test 3 edit run time: 23s

```
1 5
1 2 3 4 5
```

1 2 3 4 5 4 3 2 1
accept

test 4 edit run time: 12s

```
5 1
1
2
3
4
5
```

1 2 3 4 5 4 3 2
accept

Line 61, Column 10

Tab Size: 4 C++